

Purpose of Liftoff Reports

The United States (U.S.) Department of Energy (DOE) has published a series of reports on The Pathways to Commercial Liftoff for emerging clean energy technologies. These [Liftoff reports](#) provide a roadmap for how the public and private sector can collectively accelerate the commercialization of the technologies needed to decarbonize the U.S. economy. Given the constantly and rapidly evolving market, technology, and policy environment, these reports are designed to be “living documents” and will be updated as the commercialization outlook on each technology evolves.

Spearheaded by DOE’s Office of Technology Transitions (OTT), these Liftoff reports reinforce dialogue across not only DOE, but also other Federal departments and agencies. They build upon learnings from DOE investments and continued engagement with industry stakeholders. DOE continues to solicit input through industry forums, requests for information, and other interactions. Direct public input can be submitted via email to liftoff@hq.doe.gov.

Objectives and Scope of Virtual Power Plant Update

DOE published the [Pathways to Commercial Liftoff: Virtual Power Plants](#) report in September 2023. Since that publication, Virtual Power Plant (VPP) adoption has grown; new VPP deployments, new insights and analyses into benefits, and new tools and resources from within and outside DOE have emerged. However, deployment still needs to accelerate in the U.S. to reach 80-160 GW of VPPs (10-20% of peak load) that contribute to an affordable, reliable, and secure grid for all Americans.

This Update supplements – but does not replace – the original 2023 VPP Liftoff Report by providing additional real-world examples, new resources, and updated industry insights that support VPP deployment. This report aims to (1) communicate the differential value proposition of VPPs in meeting near-term grid challenges compared to alternatives and (2) provide proven solutions to inspire and inform near-term actions that can accelerate progress towards Liftoff.

Please reference the [2023 VPP Liftoff Report](#) for the following:

- VPP and Distributed Energy Resource (DER) definitions
- VPP value proposition
- Associated business models
- Technology in use
- Deployment potential
- Five imperatives for VPP liftoff, associated challenges, and potential solutions

Terminology

VPPs are aggregations of DERs that can balance electricity demand and supply and provide utility-scale and utility-grade grid services.ⁱ This report uses the term ‘Virtual Power Plants’ (VPPs) given it is the predominant term used in the industry, though it recognizes that other organizations use varying terms to describe similar grid assets. The National Association of Regulatory Utility Commissioners (NARUC) uses aggregated DERs (ADERs) to describe groups of DERs capable of providing one or more services to the electric grid through dispatch or control.ⁱⁱ Electric Power Research Institute (EPRI) uses the term distributed energy resource aggregations (DERAs). Other industry actors use the term distributed power plants (DPPs). This report’s definition of Virtual Power Plants includes grid assets that meet the definition of all these terms, including traditional demand response (DR).

Executive Summary

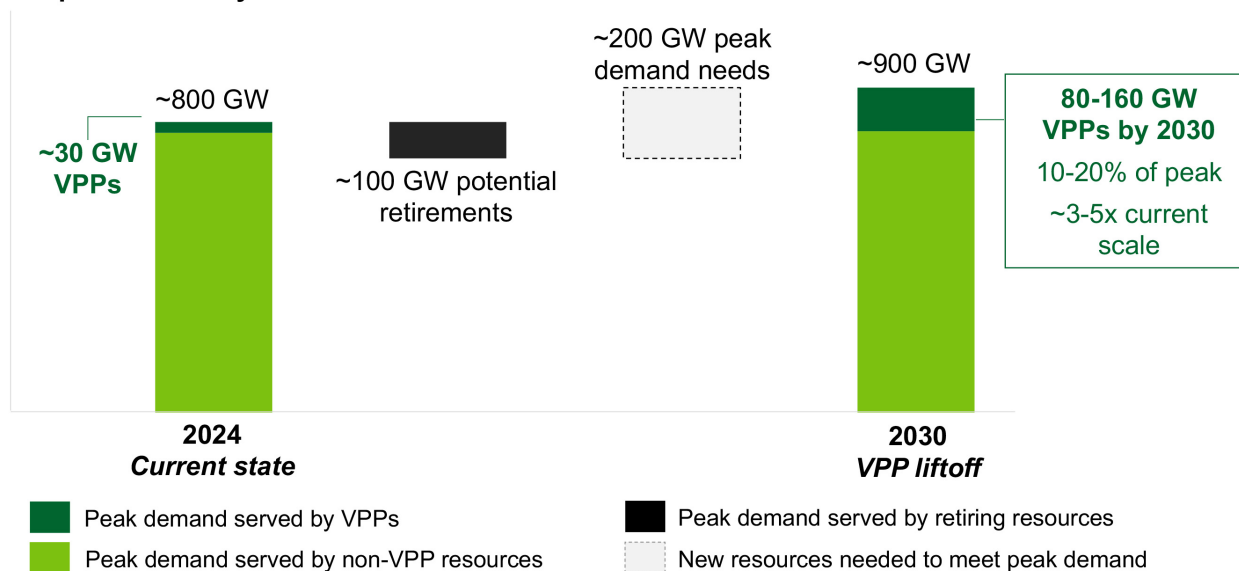
Virtual Power Plants (VPPs) are solutions that can be deployed at scale in a short timeframe to maximize the use and value of existing grid infrastructure, minimize costs to ratepayers, and ensure a resilient, reliable, and secure grid for all Americans.

Recall from the 2023 Liftoff Report

VPPs are aggregations of distributed energy resources (DERs) such as rooftop solar with behind-the-meter (BTM) batteries, electric vehicles (EVs) and electric water heaters, smart buildings and their controls, and flexible commercial and industrial (C&I) loads that can balance electricity demand and supply, as well as provide utility-scale and utility-grade grid services.

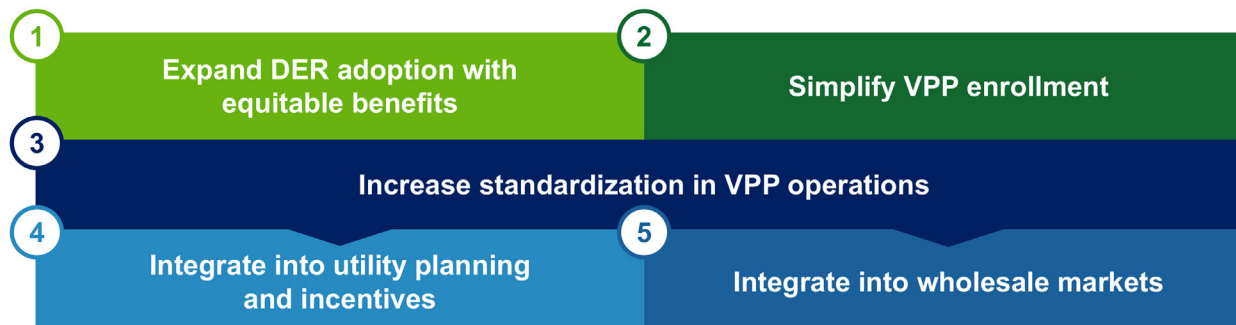
VPP liftoff

US peak electricity demand



Deploying 80-160 GW of VPPs (enough to serve 10-20% of peak load) by 2030 could support rapid load growth while reducing overall grid costs. Although VPP scale has grown over the past year to 33 GW across North Americaⁱⁱⁱ, the pace of deployment must accelerate to achieve liftoff.

Achieving liftoff will require progress on five imperatives:



Since DOE published the 2023 VPP Liftoff Report in September 2023, the pressures on the U.S. electric grid have intensified.

- **Reliability:** Peak demand is expected to increase from approximately 800 GW in 2024 to approximately 900 GW in 2030 due to growth in energy-intensive data centers, domestic manufacturing, and electrification of transport and heating.^{iv}
- **Affordability:** Utility capital investments for the transmission and distribution grid have grown by 10.8% and 14.6% respectively from 2022 to 2023.^v Capital investments are only expected to continue growing^{vi} to meet rising load growth and replace aging assets, putting upward pressure on future electricity costs for ratepayers. This increases the importance of ensuring cost-effective grid investments to mitigate cost increases for ratepayers.
- **Resilience:** The U.S. experienced a record 28 ‘billion-dollar’¹ extreme weather events in 2023 that caused \$95B of damage and injury.^{vii} These extreme weather events are responsible for 75-80% of U.S. power outages for households and businesses.^{viii}

VPPs are among the critical solutions to meet the pressing challenges the grid faces today and in the near term to keep electricity rates affordable while maintaining grid reliability and resilience.

Utilities, aggregators, policymakers, regulators, and other industry partners are taking action to implement solutions against each of the five imperatives for VPP liftoff. Replicating these proven solutions across the country could accelerate VPP deployment to reach liftoff by 2030.²



Expanding DER adoption with equitable benefits

Upfront incentives that stack across available Federal, state, city and tribal programs, inclusive utility investments, and partnerships with community-based organizations are strategies helping all communities today realize the reliability, resiliency, and affordability benefits from DERs and VPPs.

➔ For example, San Diego Community Power’s Solar Battery Savings program uses upfront, stackable incentives to provide the opportunity for no-cost solar panels and batteries for underserved communities.



Simplifying VPP enrollment

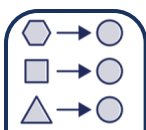
In addition to the ~30 GW of VPP capacity already enrolled today, enrolling 30-50% of the 150-200 GW of *new dispatchable* DER capacity that is projected to be added to the grid between now and 2030 could achieve liftoff nationally.

Utilities, regulators, and policymakers are harnessing existing and expected DER capacity and achieving best-in-class enrollment rates by pre-enrolling customers in VPP programs with opt-outs (instead of the opt-in method that is most common today), simplifying messaging about program benefits, and offering ongoing participation incentives.

➔ For example, Arizona Public Service’s marketplace pre-enrolls customers at point of purchase into their smart thermostat Cool Rewards program (9,290 pre-enrollments processed as of October 2024).

1 Billion-dollar events are weather and climate events that caused more than \$1B of damage.

2 For additional information on challenges and potential solutions for each of the imperatives, see Chapter 4 in the [2023 VPP Liftoff Report](#) (pages 38-52).



Increasing standardization in VPP operations

New efforts across the industry are designing standards for utility-aggregator interfaces, aggregator-DER interfaces, cybersecurity responsibilities, and other aspects of VPP operations.

Even in the absence of standards, many utilities are capturing near-term value now with basic VPP configurations that require less than \$1M in upfront investment and can be deployed in less than six months to deliver valuable peak shaving benefits. Leading utilities leverage basic VPPs as the foundation for more sophisticated models (which require enabling hardware and software) that deliver distribution grid benefits in addition to bulk system-level peaking capacity.

- ➔ *Example standardization efforts include the development of a model grid services contract from the North American Energy Standards Board and device interoperability standards from the Mercury Consortium.*
- ➔ *An example of a rapid, utility-led VPP deployment is National Grid's ConnectedSolutions program, which launched in under four months and now has 250 MW of peak shaving capacity in MA and NY.*



Integrating into utility planning and incentives

Most utilities can implement some form of VPPs today without any policy or regulatory change. However, VPP deployment has been highest in areas where state regulators and policymakers have implemented VPP-supportive actions.

Regulators are motivating utility action that is more in line with ratepayer interest by establishing cost recovery pathways for VPP-related investments, improving system planning, supporting DER deployment and aggregation, and enhancing VPP operation and compensation models. Policymakers are using legislation to accelerate deployment by establishing a direction and removing ambiguity about VPP goals and other program parameters for utility regulators and other stakeholders.

- ➔ *An example of VPP-supportive regulation is the New York Public Service Commission's Value of Distributed Energy Resources (VDER) mechanism to compensate DERs based on their system value.*
- ➔ *An example of VPP-supportive legislation is a bill signed by Colorado's legislature in May 2024, SB24-218, that requires the state's largest Investor-Owned Utility (IOU), Xcel Energy, to submit a VPP plan to the Colorado Public Utility Commission.*



Integrating into wholesale markets

CAISO and ISO-NE have fully complied with the requirements of FERC Order 2222³, theoretically unlocking wholesale market participation from a much wider range of DERs in those regions. Challenges to integrate VPPs into wholesale markets remain, particularly on data access, metering requirements, and participation models. However, market operators, state policymakers, and regulators, can collaborate to learn from each other's experiences and quickly iterate to enable VPPs to meet near-term grid capacity needs at lower costs for ratepayers.

➔ *For example, CAISO, NYISO, PJM, and SPP allow participants that meet certain criteria to use calculated telemetry readings based on sampling rather than requiring direct telemetry for each DER to participate. This allows a greater number of DERs to participate given relaxed telemetry requirements and reduced participation costs.*

Public and private sector stakeholders are taking action. This report includes over 75 examples of actions that utilities, aggregators, OEMs, regulators, policymakers, ISO/RTOs, ecosystem partners, and others are implementing today as well as **over 60 complementary programs and resources** that DOE and its collaborators have established to accelerate deployment. Stakeholders can adopt and adapt demonstrated best practices from across the country and leverage existing tools and resources to achieve VPP liftoff and contribute to a reliable, affordable, and resilient grid.

3 In September 2020, FERC (Federal Energy Regulatory Commission) approved Order 2222, which required the six FERC-jurisdictional Independent System Operators (ISOs) and Regional Transmission Organizations (RTOs) to allow participation of VPPs (referred to in the Order as "DER Aggregations") in wholesale markets. The six FERC-jurisdictional ISO/RTOs are California Independent System Operator (CAISO), Southwest Power Pool (SPP), Midcontinent Independent System Operator (MISO), New York Independent System Operator (NYISO), PJM Interconnection (PJM), and ISO New England (ISO-NE).

Introduction: Why VPPs now?

Key takeaways

- Rapid growth in peak electricity demand by 2030, capital-intensive transmission and distribution (T&D) upgrades to accommodate expected load growth, and outages due to extreme weather events and aging infrastructure are placing disproportionate pressure on grid reliability, affordability, and resilience.
- VPPs are cost-effective solutions for balancing the grid that can be deployed at scale within six months to maximize the use and value of existing infrastructure, minimize costs to ratepayers, and ensure a resilient, reliable, and secure grid for all.

Since the VPP Liftoff Report was published in September 2023, the near-term pressures on the U.S. electric grid have intensified. Forecasts of U.S. peak demand growth have increased sharply in the past year due to a surge in interest in artificial intelligence (AI) applications powered by energy-intensive data centers, hundreds of new domestic manufacturing site developments, and the continued electrification of transportation and heating. This increase in forecasted load growth will require greater utilization of local resources to satisfy electric power requirements. At the same time, recent extreme weather events have heightened awareness of the vulnerability of the grid and the need to invest in resilience. The culmination of these challenges necessitates historic investments to shore up the U.S. power system – costs that may fall on ratepayers already burdened by rising energy costs.

i. Near-term grid challenges

Reliability: *Rapid demand growth*

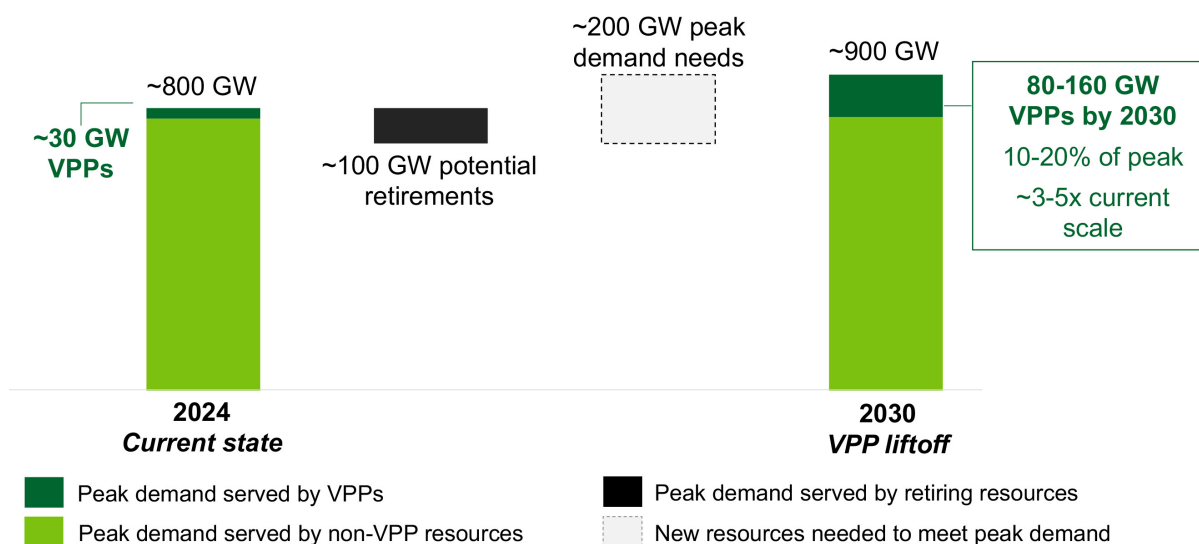
After two decades of flat electricity demand, the U.S. is returning to a period of rapid demand growth with total electricity demand expected to grow ~15-20% in the next decade.^{4,5,ix,x} The 2023 VPP Liftoff report estimated that new resources serving over 200 GW of peak demand would need to be added to the grid by 2030 to meet demand growth and replace retiring resources. Since 2023, retirement schedules and growth forecasts have both shifted, but the net result of roughly 200 GW of peak demand needs by 2030 remains.

4 NERC forecasts from December 2024 suggest total electricity will increase from 150,540 GWh in 2024 to 176,040 GWh in 2034. Total electricity demand is measured over the course of a year and is distinct from peak demand, which is a point-in-time measurement.

5 See the DOE's [Electricity Demand Growth Resource Hub](#) for additional information about and DOE resources to support rising electricity demand.

VPP liftoff

US peak electricity demand



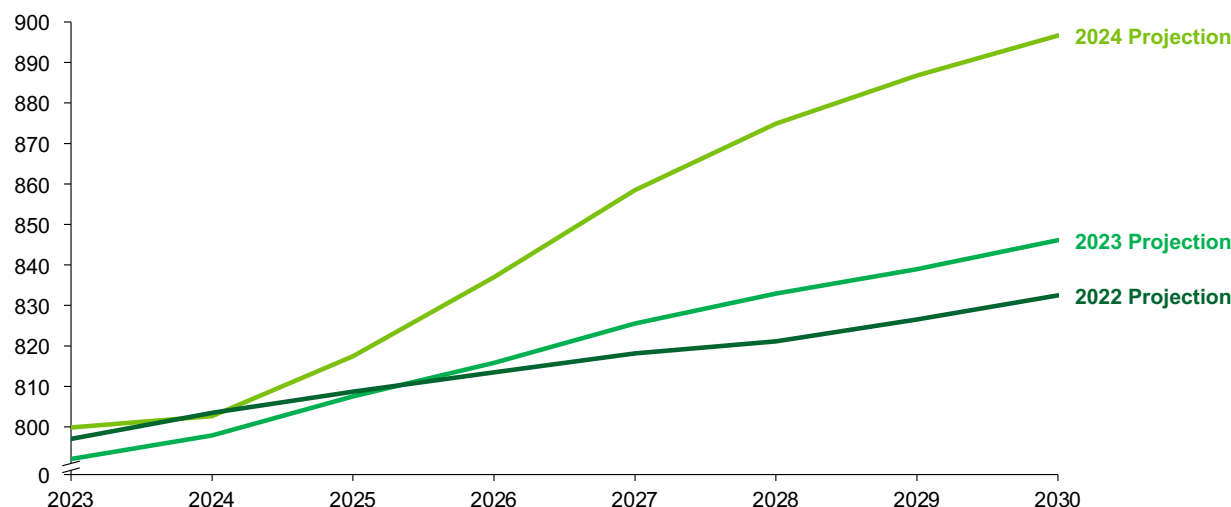
Note: NERC's Electricity Supply & Demand peak hour demand forecasts include 2024 peak summer demand as 803 GW and estimates 2030 peak summer demand to be 897 GW.^{xi,xii,xiii} NERC's 2024 Long-Term Reliability Assessment estimates that 52 GW of generators are confirmed to retire by 2029, with anticipated and announced retirements estimated to be close to 100 GW by 2030.^{xiv} For this reason, the need is estimated to be ~200 GW of firm capacity (~100 GW new peak demand + ~100 GW peak demand no longer served by retired assets, not accounting for planning reserve margin or the non-firm capacity de-rates of retiring resources). 30-60 GW estimate of VPP capacity in 2023 VPP Liftoff Report was adjusted to ~30 GW based on Wood Mackenzie's North America VPP Market Report,^{xv} which estimates that there is 33 GW of VPP capacity in North America with the majority considered to be in the U.S.

Source: NERC 2024 Long-Term Reliability Assessment, NERC 2024 Electricity Supply & Demand data, Wood Mackenzie 2024 NA VPP Market Report

Demand growth reflects economic development, though the specific drivers of demand growth vary by region. At a national level, the three primary drivers of demand growth are data center development (including to support AI applications),^{xvi} a surge in manufacturing investments (with over 900 new and expanded manufacturing facilities announced as of December 2024), and end-use electrification (e.g., transport, buildings, industrial).^{xvii,xviii}

Demand growth forecasts continue to evolve rapidly. This uncertainty increases the importance of prioritizing the most cost-effective and flexible resources to serve rapidly changing conditions.^{xix,xx}

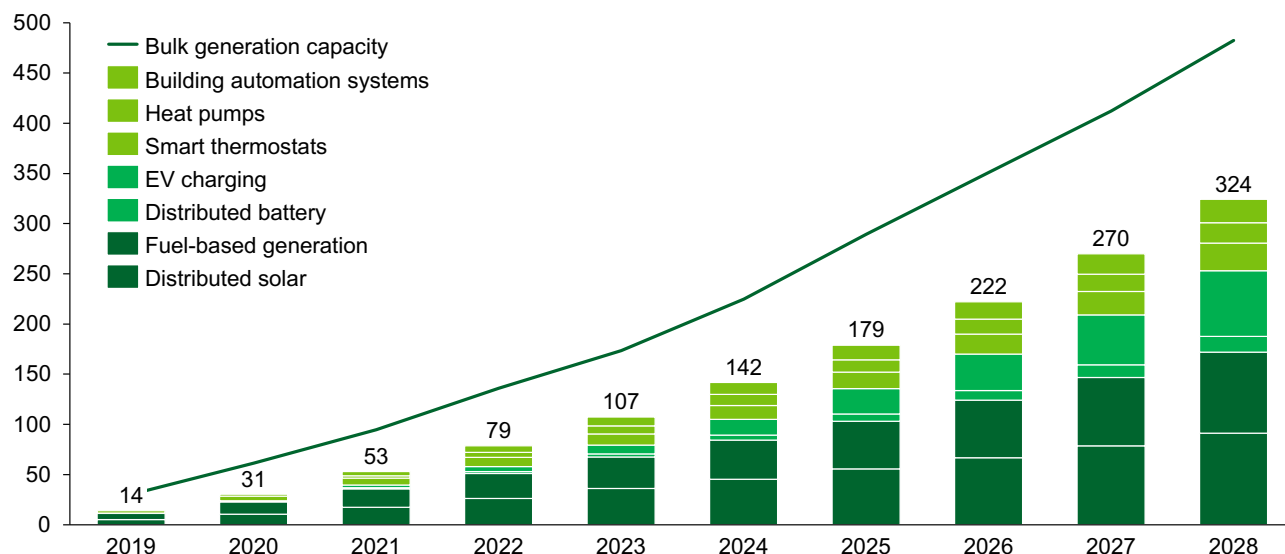
U.S. summer peak hour demand by year (2023-2030), GW



Source: NERC 2024 Electricity Supply and Demand data

Installed capacity of distributed energy resources (DERs) is forecasted to grow nearly as fast as forecasted bulk generation capacity in the next five years, with an incremental 217 GW of DERs expected by 2028.^{xxi} DER growth is expected in every state, though the pace varies regionally, with growth likely to be concentrated in specific geographies. Without efficient management of these resources, such as with a VPP, expected growth of DER capacity at the grid-edge^{6,xxii} in these regions could strain local, aging distribution systems and increase the cost to deliver electricity.

DER vs bulk generation capacity additions since 2019, GW



Source: Wood Mackenzie 2024 US DER Outlook

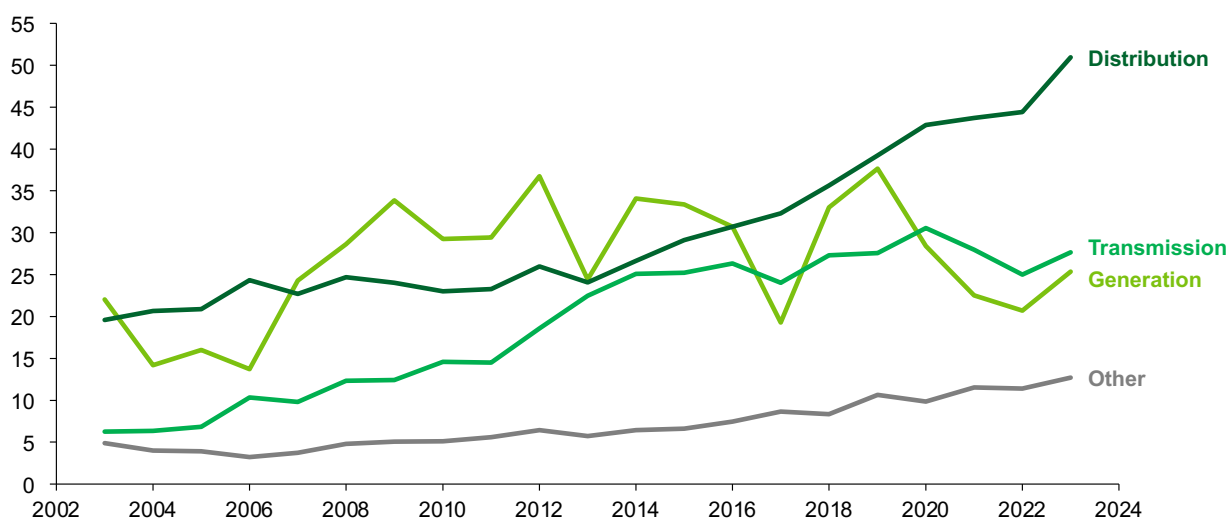
Affordability: Upward pressure on customer costs from growing capital investments

Capital investments in transmission and distribution (T&D) systems are growing to meet rising load growth and replace aging assets, impacting affordability. Over the last two decades, total utility spending on electricity generation has fallen, primarily due to lower fuel costs (e.g., growth in low-cost renewables, lower natural gas prices). However, these declines have been offset by a significant increase in T&D investment, particularly distribution investment, led by capital costs to upgrade, replace, and add new infrastructure.^{xxiii}

Utility capital investments in the distribution system grew by 14.6% from 2022 to 2023; capital expenditures (versus operating & maintenance expenditures) now comprise the majority of spending for distribution infrastructure.^{xxiv} In the U.S. Energy Information Administration's (EIA) 2023 Annual Energy Outlook (AEO 2023) projections, average combined transmission and distribution prices are expected to grow by 12% between 2023 and 2030 after accounting for inflation, even as total electricity prices decline.^{xxv} Since the release of AEO 2023, load forecasts have increased and rising load growth will further increase grid investment needs. These higher grid investments put upward pressure on future electricity costs for ratepayers.

6 The grid edge is defined as the area where the electricity distribution system transitions between the utility and the end user. Additional details are included at DOE's Supercharging the Electric Grid Edge web page.

Annual U.S. capital expenditures by sector (2003-2023), billions of 2023 U.S. dollars (\$)



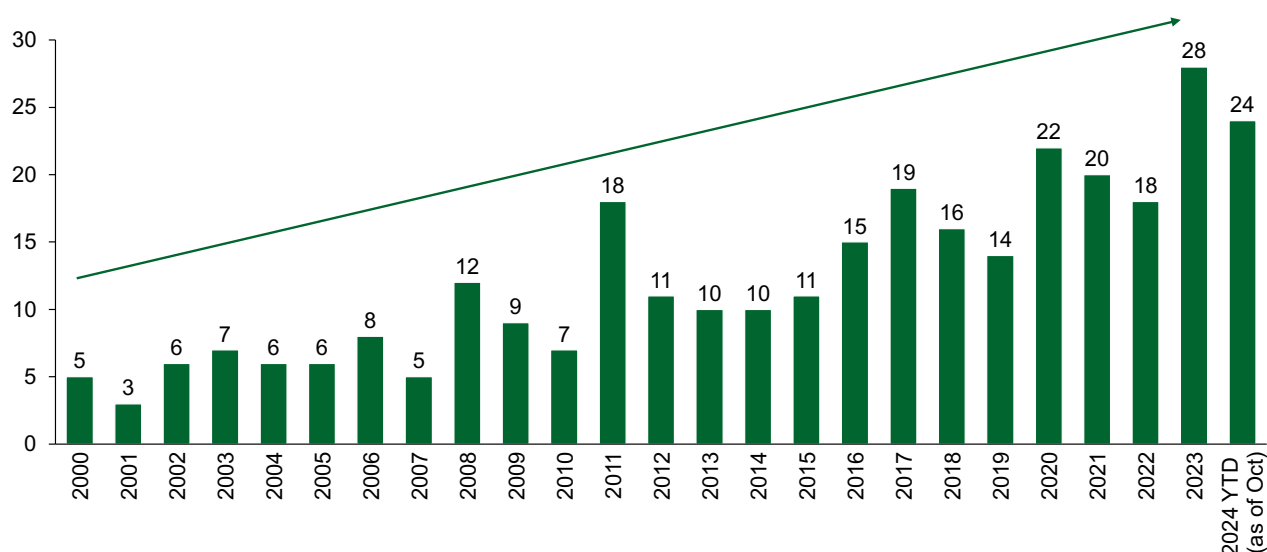
Source: FERC Form 1 (Electric Utility Annual Report)

Rising energy costs have a disproportionate impact on low-income Americans. Nearly one in four households in the U.S. was unable to pay their full energy bill for at least one month in the last year.^{xxvi} Maintaining electricity affordability in the face of increasing utility transmission and distribution investments, which have increased from 10% of customer bills in 2005 to 24% in 2020,^{xxvii} is particularly important for these households.

Resilience: Increasingly frequent extreme weather events

Increasingly frequent extreme weather threatens grid resilience. In 2023 alone, the U.S. experienced a record 28 separate billion-dollar climate disasters that caused \$95B of damage and injury.^{xxviii,7} 75-80% of U.S. power outages are due to extreme weather events, according to Climate Central.^{xxix}

Number of billion-dollar climate and weather events, count / year (adjusted for inflation)



Source: NOAA Billion-Dollar Weather and Climate Disasters

7 Climate disasters disproportionately affect underserved communities, which already often have lower energy reliability than higher income areas.

ii. VPP value proposition

How VPPs address near-term grid challenges

Near-term grid challenge	VPP value proposition	Example
 Reliability: Rapid demand growth	Impact #1: VPPs provide valuable peak shaving benefits 10-20% of 2030 peak demand could be served by VPPs	• Portland General Electric's VPP reduced peak demand load by 2% in 2024; PGE is targeting 25% of peak demand met by flexible load solutions by 2030.
	Impact #2: VPPs are quick to deploy Basic VPPs can be operationalized in <6 months to meet rapid growth	• National Grid launched its ConnectedSolutions program in under 4 months to provide peak shaving benefits.
 Affordability: Upward pressure on customer costs from growing capital investments	Impact #3: VPPs are low-cost solutions VPP peaking capacity is 40%+ cheaper than a conventional peaker plant VPPs can reduce distribution costs by providing greater locational visibility and control VPPs can offset energy bills by compensating customers	• ConEdison deferred a \$1.2B substation upgrade, spending \$200M on DERs and demand reduction measures instead. • United Power used 95 MW of flexible DER capacity and improved grid visibility to reduce transformer outages from 25,000 min/year to near 0. • San Diego Community Power uses their Solar Battery Savings Program to incentivize customers to adopt residential batteries for daily dispatch to realize \$5M of Resource Adequacy savings.
 Resilience: Increasingly frequent extreme weather events	Impact #4: VPPs improve grid reliability and resilience Solar with batteries and/or fuel generator VPPs can provide backup power during emergencies	• Duke Energy spent \$14.5M on a microgrid to provide reliable power to a rural town at a lower cost than alternatives. • Green Mountain Power's Zero Outages initiative plans to combine traditional resilience approaches with energy storage deployment through batteries and microgrids.

See 2023 [VPP Liftoff Report](#) pages 8-12 for detail on the VPP value proposition across resource adequacy, affordability, reliability & resilience, decarbonization & air pollution reduction, T&D infrastructure relief, community empowerment, and versatility & flexibility.^{xxx}

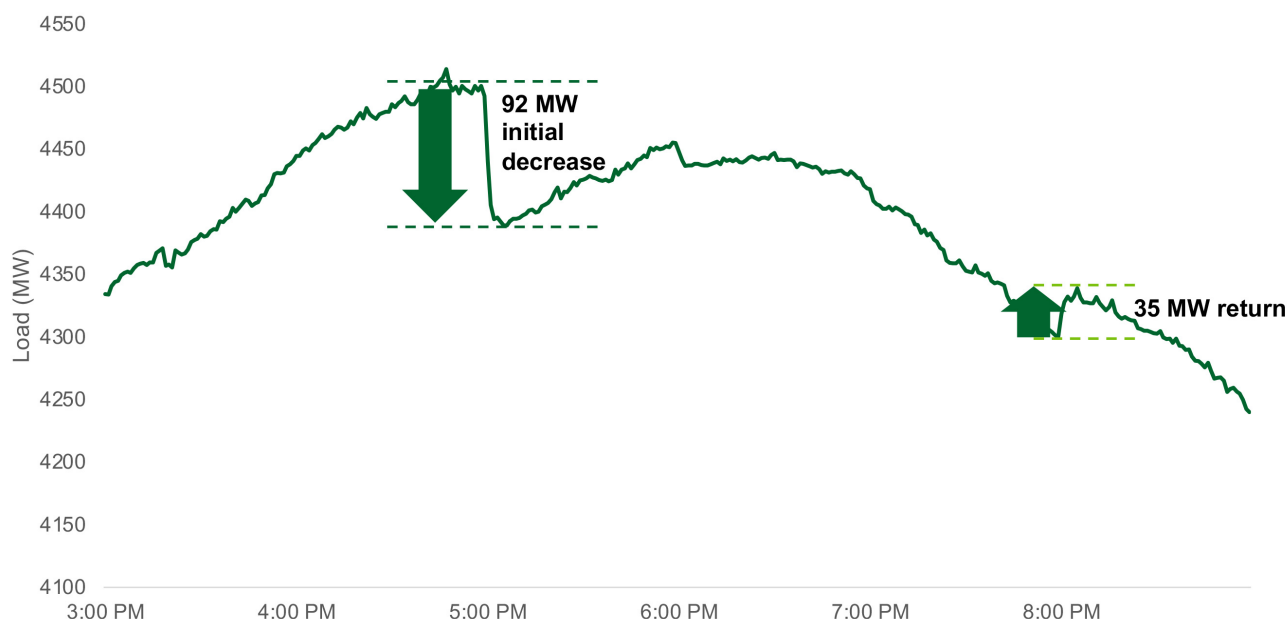
Impact #1: VPPs provide valuable peak shaving benefits

VPPs provide valuable peak shaving benefits to the grid. VPPs can aggregate DERs to serve, shift, and reduce energy demand to address growing peak demand needs and relieve grid capacity constraints. By more efficiently balancing the timing of demand with available supply, VPPs can address system constraints at the generation level (e.g., serve peak demand with storage DERs), at the transmission level (e.g., reduce peak demand when utility-scale supply is limited by transmission constraints), and distribution level (e.g., shift peak demand that threatens to exceed the safety limits of local equipment to earlier or later in the day).

One example of a utility taking advantage of this potential is **Portland General Electric (PGE)**. PGE plans to grow its VPP from serving ~2% of peak demand today to ~25% of peak in 2030.^{8,xxxix,xxxii} PGE plans to increase its VPP capacity by encouraging greater participation from new and existing solar and storage assets, flexible customer loads, and customer back-up generation.

8 PGE has been growing their Customer Flexible Load programs and VPP capabilities for over two decades.

Bulk system impact from Portland General Electric's peak shaving program (August 14, 2023)



Source: Portland General Electric

Impact #2: VPPs are quick to deploy

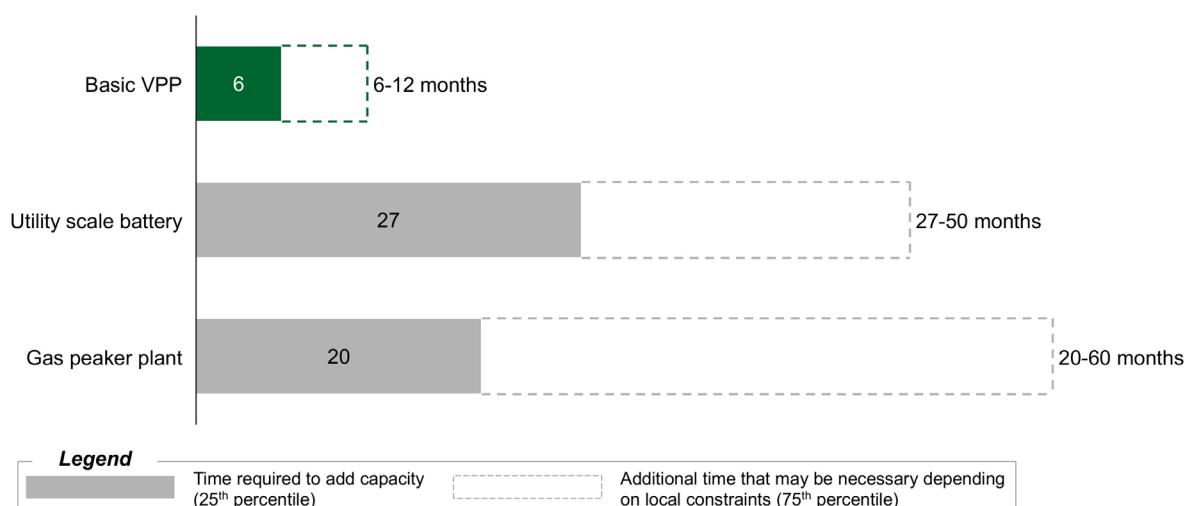
Basic VPPs that shave demand peaks can start operating within six months^{xxxiii}; this can buy time for the construction of higher-capacity assets and increase the value of grid assets for which Americans have already paid. Basic VPP configurations⁹ can leverage DERs that are *already on the grid* or expected to be deployed to serve as the foundation for fast-launching, large-scale VPP programs. Wood Mackenzie estimates that U.S. consumers and businesses will install 324 GW of new DERs between 2019-2028, representing 137 GW of curtailable capacity¹⁰ if enrolled in VPPs.^{xxxiv}

Traditional approaches to increasing grid capacity (utility-scale generation, transmission, distribution) rely on investing in large, centralized physical infrastructure, such as building fossil fuel-powered peaker plants and upgrading transformers. These upgrades are facing lengthening delays for several reasons. New electricity generation facilities are waiting four to six years in transmission interconnection queues before they can connect to the grid to supply power.^{xxxv} Long distance greenfield transmission projects often face lengthy permitting timelines, with review periods that average 4.3 years and can extend up to 11 years.^{xxxvi} Lead times to procure large transformers (greater than 500 MVA) are averaging three years due to supply chain issues.^{xxxvii}

⁹ For an explanation of basic vs. more sophisticated VPP configurations, reference Chapter 3: Increasing standardization in VPP operations.

¹⁰ Curtailable capacity includes flexible capacity from smart thermostats, heat pumps, buildings with energy management systems, and export potential from batteries.

Timeline to add 20 MW of dispatchable peaking capacity, months



Note: Industry participant interviews informed the timeline for basic VPPs, supported by RMI's Reliability Brief from July 2024^{xxxviii}. For utility scale battery and gas peaker plant, the timeline includes time from interconnection request to project Commercial Operations (COD) for projects with 2017-2023 CODs; displaying 25th to 75th percentile range. Median values are 40 months for battery and 42 months for gas projects.^{xxxix}

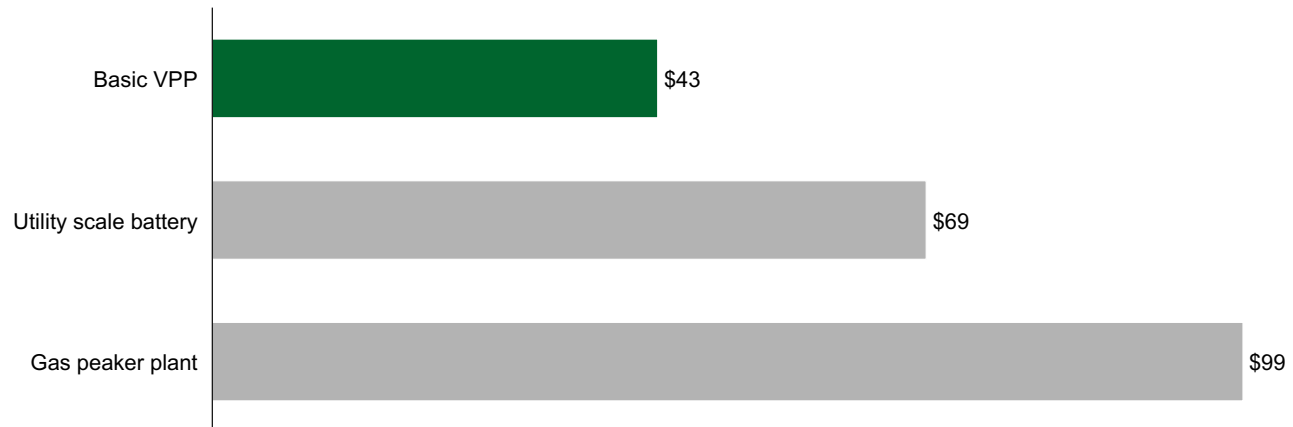
Source: Industry interviews, RMI Reliability Brief, Lawrence-Berkeley National Lab's 2024 Queued Up Report

As an example of how VPPs can address rising electricity demand, **NRG Energy** and **Renew Home** recently announced plans to launch a 1 GW VPP in Texas by 2035, equivalent to 12 gas-fired peaker plants, by leveraging flexible demand from smart thermostats. This announcement comes just months after the Electricity Reliability Council of Texas (ERCOT) revised its 2030 load growth forecasts to 148 GW^{xi}, **an increase of 40 GW from last year's forecast.**^{xli} Rapid peak demand growth requires solutions that can be deployed rapidly.

Impact #3: VPPs are low-cost solutions

VPPs are a cost-effective peak capacity resource relative to traditional investments, both on the bulk power grid and the distribution grid. On the bulk power grid, procuring new system-level peak capacity from a VPP can be lower cost than procuring the same capacity from a natural gas peaker plant or utility-scale battery. These savings, as well as reduced distribution and transmission costs, accrue to all ratepayers (not just VPP participants). An RMI study of an example utility system in 2035 found that a VPP-enabled portfolio reduces net power generation costs by 20% or roughly \$140 per household (including non-participating ratepayers) per year compared to a baseline scenario.^{xlii} In New York, **ConEdison** deferred a \$1.2B substation upgrade in 2014, spending \$200M instead on DERs and demand reduction measures as part of the Brooklyn Queens Demand Management Program.^{xliii} Beyond these system benefits for all ratepayers, additional financial benefits accrue to customers enrolled in the VPP in the form of incentive payments.

Comparison of net cost to an example utility of providing 400 MW resource adequacy across three options, Net cost per kW-year



Note: Values for 400 MW of peaking capacity are shown in \$/kW-yr. The VPP analyzed consists of smart thermostats, smart water heating, home EV managed charging, and BTM battery demand response. Modeled equipment subsidy costs to utility are \$75 for smart thermostats, \$315 for smart water heaters, and \$0 for EV charging and BTM batteries. Marketing costs assumed at \$50 per device. Utility studied is assumed to have 50% renewable generation mix, with resource adequacy needs in summer and winter. 8760 hours were considered, and resources must be able to operate in 63 peak hours (when top 400 MW are needed) spanning 7 months, for 7 consecutive hours at a time. Benefits of emissions reduction and resilience are not shown; when included, VPP net cost is lower, though actual emissions impact will vary by local grid mix.^{xliv}

Source: Brattle Group's Real Reliability: The Value of Virtual Power Report

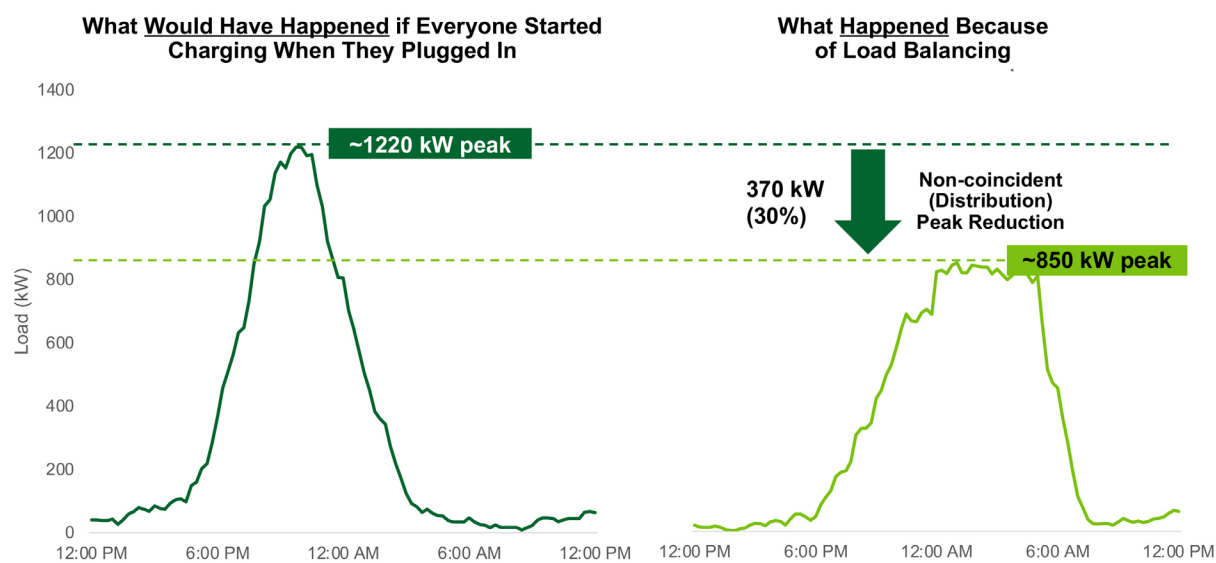
On the distribution grid, VPPs can help utilities defer costly system upgrades by shaving localized peak loads that would otherwise trigger the installation of new equipment. Many utilities facing rising demand are using VPPs as an interim solution until distribution grid capacity upgrades are absolutely necessary, saving ratepayers money in the meantime. A whitepaper co-authored by **AES Indiana** and **Camus Energy** found that deploying visibility solutions to detect where EVs are located on the grid and transitioning to grid-optimized managed charging programs could defer upgrades to 85% of eligible feeders and service transformers for an average of 8.5 years compared to a business-as-usual scenario. Capital cost savings from deferring upgrades were estimated to avoid close to \$1B in cost overruns over the next decade, with savings going directly to consumers.^{xlv} AES Indiana is *one* utility with 500,000 utility meters. While service areas across the U.S. are diverse, extrapolating to the 150 million meters across the U.S. would imply **significant potential savings¹¹ by deferring capital investments and optimizing the use of the existing electricity system across generation, transmission, and distribution.**^{12,xlvi}

Baltimore Gas & Electric's (BGE) managed charging program demonstrates the peak shaving potential of VPPs on the distribution grid. With a feeder-level participant group of 880 vehicles, BGE's managed charging program created a non-coincident peak reduction of 30% while still serving customer's transportation energy needs. BGE plans to grow its managed charging program from 3,253 residential customers to 30,000 customers by 2027.^{xlvii,xlviii}

¹¹ A rough extrapolation of this example to the 150 million meters across the U.S. would imply potential savings of \$300 billion over the next decade by deferring capital investments.

¹² An [LBNL study mentioned the U.S. building sector alone could avoid over \\$100B per year](#) in power sector costs by leveraging demand-side solutions (e.g., smart thermostats, electric heat pumps, smart control systems) by 2050.

Distribution grid impact from Baltimore Gas & Electric's managed charging program



Source: WeaveGrid

Impact #4: VPPs improve grid reliability and resilience

VPPs provide resilience benefits that traditional generation assets cannot provide—and at a lower cost than alternatives. VPPs that include solar and storage or fuel generators at a household or commercial and industrial site provide power with far fewer possible points of failure than power supplied from a distance by a traditional power plant. VPPs also have the potential to help utilities restore power to impacted areas more quickly, reducing the length of outages for customers impacted by severe weather events.

Much of the grid hardening work in disaster-prone areas has been undergrounding power lines. Although this has been effective in some areas, including pockets of Florida during Hurricanes Helene and Milton^{xlix}, it has come at a high cost. The **Public Service Commission of Wisconsin** estimates that undergrounding a 69-kilovolt line costs ~5x more per mile versus aboveground installation.ⁱ Alternatively, utilities are using DERs and VPPs at the end of vulnerable transmission or distribution lines to ensure reliable power at a lower cost than undergrounding lines.

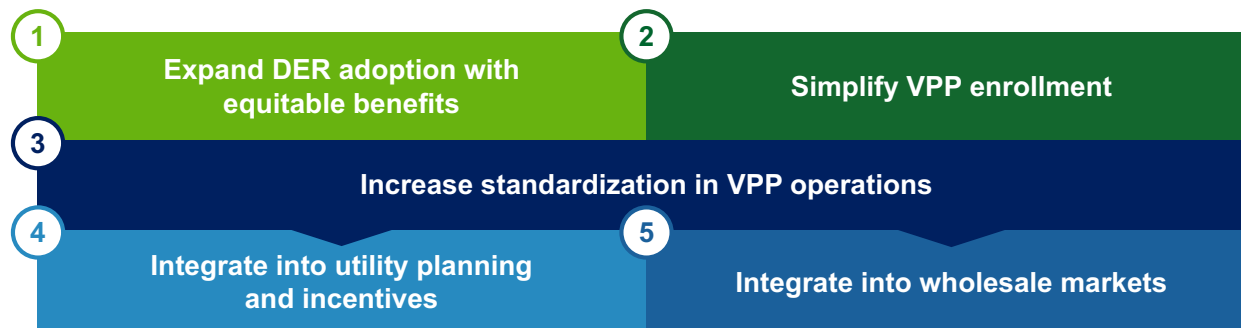
For example, in 2023, **Duke Energy** installed 2 MW of solar power and 4.4 MWh of battery storage along with VPP-enabling technology to create a microgrid in Hot Springs, a town of about 535 residents in North Carolina. With a cost of \$14.5M, the microgrid was deemed less expensive than the grid upgrades that would have been required to provide reliable power for the rural town. For the residents of Hot Springs, the microgrid offered resilience; for the surrounding communities, it provided energy and bulk system benefits, such as frequency and voltage regulation and capacity during system peaks.^{li} With the Hot Springs microgrid and VPP-enabling investments, Duke Energy restored power to residents quickly after Hurricane Helene in 2024, even though the local substation was severely impacted by flooding.^{lii}

iii. Imperatives for VPP liftoff

VPPs are solutions that can be deployed at scale in a short timeframe to maximize the use and value of existing grid infrastructure, minimize costs to all ratepayers, and ensure a resilient, reliable, and secure grid for all Americans.

Deploying 80-160 GW of VPPs (enough to serve 10-20% of peak load) by 2030 could support rapid load growth while reducing overall grid costs. VPPs are not new and have been operating with commercially available technology for years.^{liii} While VPP scale has grown over the past year to 33 GW across North America^{liv}, deployment must accelerate to achieve liftoff by 2030.

As explained in the 2023 VPP Liftoff Report, achieving liftoff for VPPs will require progress on five imperatives:



Utilities, policymakers, regulators, and other industry partners all have a role to play in accelerating action against these five imperatives to address the challenges hampering VPP adoption today.

The potential for VPPs to meet near-term grid needs cost-effectively for American ratepayers represents an urgent call to action for all grid stakeholders to do their part in advancing deployment.

Building on the foundation of the 2023 VPP Liftoff Report, the remainder of this Update will explore each of the five imperatives. Starting with a brief overview of the imperative, each chapter and its corresponding appendix will focus on presenting new VPP case studies, new insights into VPP benefits, and new tools and resources from the Department of Energy and broader industry that can support power sector decisionmakers and accelerate progress towards VPP liftoff.

Chapter Four: Integrating into utility planning & incentives

Key takeaways

- Across the U.S., VPP deployment has been highest in the states with supportive state regulatory and/or policy actions.
- Many state utility regulators – public utilities commissions (PUCs) and public service commissions (PSCs) – have opened regulatory proceedings within the last 18 months to advance VPP adoption. Examples include requiring longer-term distribution grid planning that incorporates consideration of VPPs and establishing or revising compensation mechanisms to better align utility financial incentives to positive grid and customer outcomes.
- Legislative changes to utility regulations or policy are not necessary for investor-owned utilities to deploy VPPs today, but can accelerate deployment by establishing a direction and removing ambiguity about VPP goals and other program parameters (e.g., types of DERs, desired grid services). Examples include Colorado and Maryland legislative actions.
- Regulators and policymakers approaching VPPs today can draw from the menu of 22 policy actions underway across the U.S. to inform program design and integrate VPPs into utility planning and incentives.

Note: This chapter discusses state regulatory and policy actions that are most relevant for investor-owned utilities (IOUs) regulated by state PUCs/PSCs. Governing bodies of other utilities (e.g., member boards of co-ops, city councils overseeing public power, tribal utility authorities) can also look to these levers for consideration, but the historical financial disincentives impacting IOUs may be less relevant to nonprofit cooperatives and municipally run utilities.

4.i. Utility financial incentives and VPP deployment

In the era of flat electricity demand over the last two decades, VPP deployment by investor-owned utilities (IOUs) was in part stifled by a lack of financial incentives because it meant lower utility profits. Under conventional regulatory models, IOUs can earn an authorized return on equity (typically 9-11% annually) on capital investments; thus, IOUs deploying a low-capex VPP to add system capacity instead of a traditional capex-heavy investment (e.g., a peaker plant) would have realized lower profits.⁴⁵

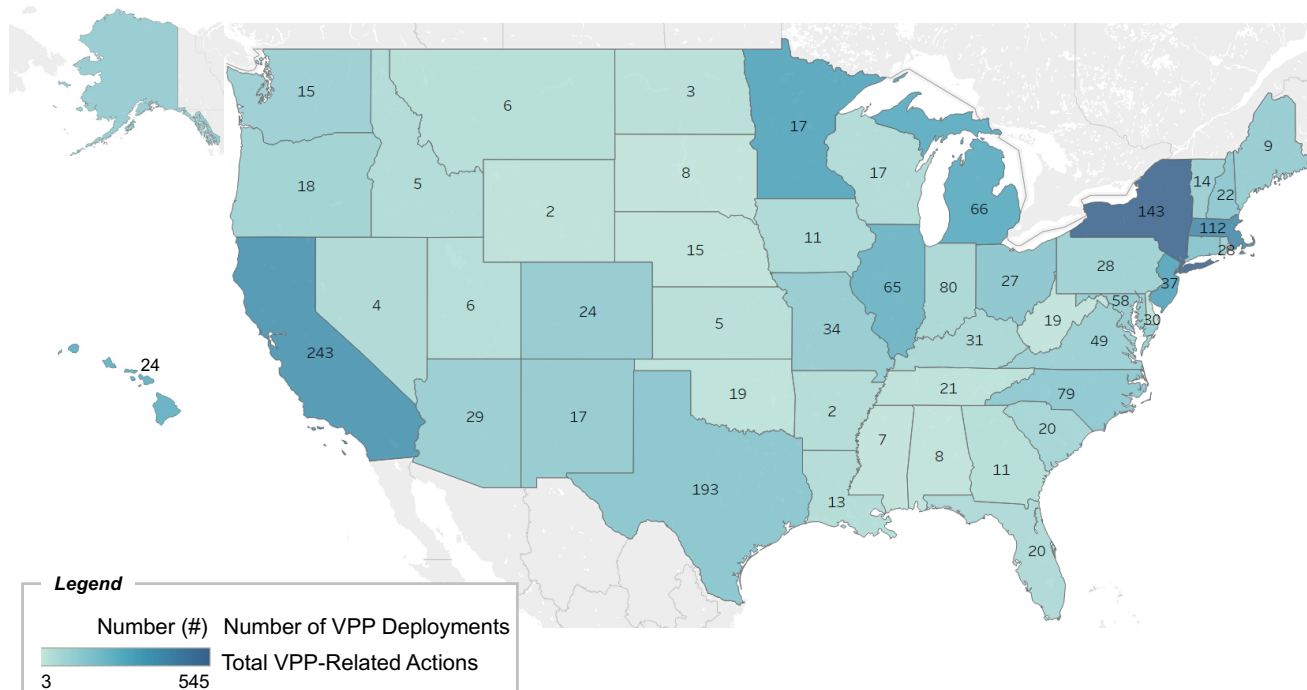
Today, rising electricity demand and the need to replace aging grid infrastructure means many utilities have rapidly growing capital needs. In this context, IOUs can deploy VPPs to help meet system needs and interconnect more load,^{46,xciv,xcv} while creating room in their budgets for necessary capital-intensive investments elsewhere (e.g., transmission expansion, new bulk power generation assets). Additionally, state regulators and policymakers are applying pressure to limit capital expenditure and increasingly pushing back on utility investment plans to ensure any increase in customer bills is fully warranted. For example, in December 2023, the **Illinois Commerce Commission** rejected **Ameren Illinois** and **Commonwealth Edison**'s multi-year integrated grid plans over concerns that the utilities did not adequately "consider affordability and cost-effectiveness [criteria] so that customers are not unfairly asked to shoulder undue costs."^{xcvi}

⁴⁵ See page 48 of the [2023 VPP Liftoff](#) report for additional detail on utility compensation structures.

⁴⁶ For example, since 2017, Arizona Public Service (APS) has been developing demand response and load management programs (with approval from the Arizona PUC) to aggregate DERs into VPP programs, helping manage growing load (as discussed in APS' [2019 IRP Draft](#)). In Minnesota, Northern States Power Company (doing business as Xcel Energy) introduced the concept of distributed capacity procurement (DCP) in [comments](#) for an IRP filing (Docket No. E002/RP-24-67) and said the program could provide 400 – 1,000 MW of capacity; actual plans with specifics would be included in the future IRP.

While IOUs can start implementing VPPs without any regulatory or policy changes, supportive regulatory and policy action is accelerating VPP deployment. Across the U.S., localities where state regulators or policymakers have taken VPP-supportive actions have seen the highest number of total VPP deployments to date.

Number of VPP deployments (as of July 2024) vs. state policy/regulatory VPP-related actions (2020-Q3 2024), count



Note: Number of VPP deployments based on Wood Mackenzie data as of July 2024.^{xvii} Wood Mackenzie defines a VPP deployment as: "The association of a vendor aggregation and a DER program. Aggregation is broadly defined to consist of DERs or loads directly under vendor management, or under the management of a downstream device partner. Example: If three vendors partner on a VPP that is monetized through two programs, there will be six deployments recognized." State regulatory and policy actions based on North Carolina Clean Energy Technology Center and includes data from Q1 2020 – Q3 2024.^{xviii} VPP-related state policy/regulatory actions include all types of actions tracked by DSIRE Insight (studies, policy, incentives, deployment, rates) that include the technology tag: demand response, grid modernization, smart grid, storage, AMI, DER, distribution system planning, data access, VPPs.

Source: Wood Mackenzie 2024 NA VPP Market Report, North Carolina Clean Energy Technology Center Policy & Regulatory Actions

State regulators and policymakers play a critical role in enabling a statewide VPP approach to support easier scale up across utility jurisdictions. While several utilities have pursued VPP deployments before any policy or regulatory action, state policy and regulatory efforts have been important to supporting broader adoption by integrating VPPs into standard utility processes (including planning and cost recovery) and aligning utility and ratepayer incentives. As discussed in the 2023 VPP Liftoff Report, increased VPP standardization will accelerate VPP integration in utility planning and incentives.⁴⁷

47 See page 35 of the [2023 VPP Liftoff Report](#) for additional discussion on the imperatives.

4.ii. Supportive regulator actions for integration into utility planning and incentives

All state regulators have the authority to pursue actions that could support VPP deployment.⁴⁸ As part of their mandate to ensure affordable and reliable electricity service, PUC/PSCs can proactively direct utilities to fairly consider VPPs alongside ongoing conventional capital investments (e.g., bulk power generation, transmission, distribution) to meet grid needs. In many states, PUCs' legacy organizational models, limited staff capacity, and reactive cultures have resulted in limited proactive engagement with utilities before they submit investment plans (e.g., providing proactive guidance on considering VPPs).^{xcix} This is starting to change as mounting load growth, affordability, and reliability pressures on the grid are motivating several state PUCs to proactively provide direction and establish programs that can influence IOU investments, including VPP deployments.

Supportive regulator actions for VPP integration into utility planning and incentives (*not exhaustive*)

 Utility cost recovery	 System planning	 DER deployment	 DER aggregation	 VPP operations
Establishing utility cost recovery methods for VPP-related investments	Improving grid planning processes to better integrate VPPs as a solution	Implementing or revising programs to increase DER deployments, which enhance VPP potential	Developing DER aggregation models and deployment requirements to enable VPPs	Supporting VPP operations to proactively address common VPP deployment barriers
• Massachusetts • Michigan • Vermont	• Georgia • Massachusetts • Minnesota	• Colorado • Michigan • New York • South Carolina	• California • Colorado • Texas	• Connecticut • Massachusetts • New York • Rhode Island

See [Appendix D.i](#) for a menu of additional regulatory and policy options that regulators and policymakers can consider alongside case studies of how those options have been implemented to date.

Example types of actions that PUCs have pursued recently to drive uptake of VPPs include:

1. Utility cost recovery: Establishing utility cost recovery methods for VPP-related investments.

PUCs are implementing performance incentive mechanisms (PIMs)^{49,c} and clarifying what types of VPP-related investments are eligible for cost recovery (e.g., including DERs in utility rate base,^{50,ci} capitalizing software costs).

2. System planning: Improving grid planning processes (e.g. integrated distribution system planning) to incorporate VPPs as a solution.

PUCs can implement integrated distribution system planning to require objective-driven planning (e.g., grid reliability improvements, customer empowerment), planning over long-term time horizons (e.g., 10+ years), and consideration of comprehensive solutions to address grid needs (e.g., DERs, VPPs).^{51,cii} Today, 21 states and the District of Columbia require utilities to file integrated distribution system plans^{ciii}.

48 State regulators include public utility and public service commissions (PUC/PSCs) that regulate most investor-owned utility (IOU) planning, operations, and retail compensation as relates to VPP deployment and the distribution system. PUC/PSCs review and approve IOU capital investment plans and have authority over determining IOU capital return rates, customer rate designs, and other distribution system plans. See VPP Liftoff report page 20 of the [2023 VPP Liftoff Report](#) for additional info on state regulator roles and responsibilities.

49 Performance incentive mechanisms (PIMs) is a type of performance-based regulation (PBR). PBR is a regulatory model that financially rewards utilities for positive ratepayer outcomes, rather than returns on capital expenditures. See NARUC's [PBR Overview](#) to learn more. See RMI's [PIM Database](#) for a longer suite of examples of PIMs that PUCs and utilities have implemented.

50 Proposals for utilities to own behind-the-meter devices, such as batteries or generators, [have drawn criticism](#) from some industry participants who say that although such an approach may lead to faster scale-up of VPPs, it may lead to higher electricity prices for customers because of the utility's monopoly power, lack of competition to drive down prices, and its guaranteed financial return on the devices when included in the rate-base.

51 See Lawrence Berkeley National Lab's [Integrated Distribution System Planning](#) resource hub, which includes an interactive planning framework, a map and detail on existing planning requirements by state, and information on technical assistance and resources available to PUCs and policymakers.

broader use of integrated distribution system planning practices could promote a more proactive utility investment approach to better consider and use DERs and VPPs. Regulatory approaches requiring or directing utilities to invest in grid orchestration platforms (e.g., DERMS) as part of broader grid modernization efforts can help distribution utilities manage an increasingly complex and digital grid, which also establishes the technology foundation for future VPP deployments.⁵²

3. **DER deployment: Implementing or revising programs to increase DER deployments, which enhance VPP potential.** PUCs are studying and establishing methods to increase DER deployments, such as streamlining interconnection processes, establishing customer incentives, and testing pay-for-performance compensation mechanisms and DER-supportive tariffs. DERs can support net cost savings for customers and increase the resources available for DER aggregation.
4. **DER aggregation: Developing DER aggregation models and deployment requirements to enable VPPs.** PUCs are requiring utilities to develop pilots or consider how to aggregate DERs into a VPP program to be used as a grid asset.
5. **VPP operations: Supporting VPP operations to proactively address common VPP deployment barriers.** PUCs are increasingly influencing VPP operations to maximize system value, including by engaging on data access challenges⁵³ and establishing tariff structures that better compensate VPPs for their full suite of grid benefits (e.g., capacity, reliability, decarbonization impacts, etc.) and enabling value stacking (including stacking across both retail and wholesale market revenue streams).

Specific state examples illustrate how PUCs are putting these types of actions into practice.

Case Study: Colorado PUC, CO

Colorado PUC established a performance incentive mechanism to accelerate DER interconnection, helping improve DER deployment to support VPP potential.



- ▶ **Colorado PUC** approved a performance incentive mechanism for **Xcel Energy** to speed up interconnection of DERs ([Order 23AL-0188E](#)) in October 2023.^{civ}
- ▶ The PIM requires Xcel to refund customers 4% of the interconnection fee per day delayed beyond Xcel's internal timeline targets (e.g., 50 days).
- ▶ If Xcel interconnects the DER faster than the target timeline, the value would be credited against any penalties accrued for exceeding the target.
- ▶ The PIM aims to align Xcel incentives with ratepayer interests to support DER interconnection, enabling faster DER deployment and supporting greater VPP potential at scale.

52 See DOE's [Innovative Grid Deployment](#) Liftoff report for additional information on other grid modernization technologies and foundational platforms available to support modernizing distribution grids.

53 See [Chapter 5.iv](#) for additional detail on VPP-related data access challenges and potential solutions.

Case Study: New York State PSC, NY

New York State PSC implemented a value compensation methodology to reward DERs for a range of delivered grid benefits.



- ▶ In 2017, **New York State PSC** implemented a [Value of Distributed Energy Resources](#) Value Stack (VDER, or the Value Stack) to better compensate and incentivize DERs for provided grid value.
- ▶ The Value Stack includes six values to determine DER compensation:
 - » Energy Value (Locational Based Marginal Price, LBMP)
 - » Capacity Value (Installed Capacity, ICAP)
 - » Environmental Value (E)
 - » Demand Reduction Value (DRV)
 - » Locational System Relief Value (LSRV)
 - » Community Credit (CC)
- ▶ This model allows for value stacking across multiple revenue streams (including wholesale market revenues) to fully reward DERs for delivered grid benefits.
- ▶ The Value Stack provides location-specific compensation to reward VPPs that have the greatest impact on alleviating distribution system constraints.

Detailed case study provided in [Appendix D.ii](#).

State by state, PUC/PSCs have different policy contexts and starting points of regulatory frameworks that can be used – or adjusted – to encourage VPP deployment.⁵⁴ When motivated to support VPP deployment, PUC/PSCs can leverage components of the real-world examples described above to tailor regulatory actions that are appropriate for their state's context and grid objectives.

Regulators have reported success with directing a few staff members to develop simple VPP regulatory frameworks (e.g., a smart thermostat program) and then adding resources and scaling up over time towards more complex regulatory efforts as impacts are proven out and lessons are learned.^{cv}

Regulatory approaches will likely continue evolving over time as VPP program design and underlying technology also evolve. To enable continuous improvement, PUCs could consider establishing processes that enable and encourage evolution. For example, the **Hawaii PUC** built in iteration to revisit elements of DER programs (e.g., incentive levels, operational characteristics) every three years with stakeholders to keep pace with an evolving grid.⁵⁵ The **Connecticut Public Utilities Regulatory Authority (PURA)** implemented a "regulatory sandbox" program that fosters new grid technology deployments and informs enabling regulation.^{56, cvi}

⁵⁴ For policymakers/regulators considering implementing a VPP initiative, RMI/VP3 defined a set of guiding policy principles that can help inform initial actions to maximize long-term benefit (See [Appendix D.v](#) for the full set of policy principles).

⁵⁵ See additional detail about Hawaii's DER program evolution in NARUC's [Aggregated DER in 2024: The Fundamentals](#) (page 69).

⁵⁶ Connecticut PURA established the [Innovative Energy Solutions Program](#) in 2023 to encourage grid innovation, including defining features such as a four phase process from ideation to scale up, cost recovery guidance, and screening and performance metrics.

4.iii. Supportive policymaker actions for integration into utility planning and incentives

Legislative changes to utility regulations or related policy are often not *necessary* for investor-owned utilities to deploy VPPs or for regulators to take action, but they can be an *accelerant*. Legislation and other policy measures can shorten design and deployment timelines by removing ambiguity about VPP goals and other program parameters or aligning expectations with state energy and climate goals.

At the state level, policymakers (e.g., legislators, governors, tribal governments) can empower PUC/PSCs in states where regulators may not consider it their role to proactively shape VPP programs and/or the processes underpinning their deployment (e.g., filing dockets, RFIs, etc.). In these states, policymakers can accelerate regulatory processes, potentially by years, by providing direction and focus while giving PUCs and utilities room to determine the most effective regulatory frameworks. Similarly, tribal governments can also provide direction to tribal utilities to advance VPP-supportive actions. In Colorado, Massachusetts, and New York, actions by policymakers built on previous PUC actions to strengthen and provide explicit support to grid modernization and VPP supportive efforts.

Three types of actions that state policymakers have recently taken to support VPP deployments include:

1. **Establishing grid modernization policies and VPP-enabling requirements to enhance system planning:** **Washington State** passed [HB 1589](#) in March 2024 that required utilities to submit integrated system plans. VPP-enabling features include requiring plans to align with state clean energy goals and emission reduction targets.
2. **Requiring utilities and PUCs to develop VPP programs and/or supportive tariff mechanisms:** **Colorado** passed [SB24-218](#) in May 2024 that requires the state's largest IOU (Xcel) to submit a VPP plan to the PUC. This built on ongoing actions by the **Colorado PUC** to advance VPP programs as part of an effort to serve rising demand while mitigating costs for ratepayers.
3. **Clarifying VPP stakeholder roles and requirements:** **Texas** legislators passed [SB 1699](#) to establish third-party aggregation requirements for DERs and to authorize the **TX PUC** to establish rules and requirements for DER aggregators.

See [Appendix D](#) for a menu of 22 regulatory and policy options to support VPPs ([Appendix D.i.](#)), detailed case studies on New York PSC's Value of DER (VDER) Value Stack compensation method and Massachusetts legislation on grid modernization planning requirements ([Appendix D.ii.](#)), 6 additional resources ([Appendix D.iii.](#)), 9 supportive DOE programs ([Appendix D.iv.](#)), VPP policy principles from the Virtual Power Plant Partnership (VP3) ([Appendix D.v.](#)), and a summary of existing benefit-cost assessment frameworks available to support VPPs from NARUC ([Appendix D.vi.](#)).

Chapter Five: Integrating into wholesale markets

Key takeaways

- In the last decade, wholesale markets have been the primary mechanism to provide and monetize grid services from distributed flexible loads – particularly commercial and industrial loads.
- FERC Order 2222 has the potential to unlock wholesale market participation from an enormous amount of DER capacity. At a time when capacity markets are tight (e.g., PJM), VPP participation in wholesale markets has never been more important for system affordability and reliability.
- Although industry actors have been excited about the potential impact of Order 2222, slow implementation timelines, varied approaches across ISO/RTOs, and obstructive state, ISO/RTO, and utility rules have blocked the full integration of VPPs into wholesale markets.
- Technology, regulatory, and policy solutions are emerging domestically and internationally to remove barriers for VPP integration into wholesale markets. Industry collaboration is needed to share learnings and accelerate implementation.

5.i. VPP wholesale market participation today

In the last decade, wholesale markets have been the primary mechanism to provide and monetize grid services through demand response for distributed loads – particularly commercial and industrial loads. Today, 29 GW of demand response participates in wholesale markets.^{cvii}

All seven of the U.S. ISO/RTOs allow wholesale market participation from VPPs that manage demand without exporting power to the grid. Well-established demand response aggregators such as **CPower** continue to focus their business strategy on wholesale markets, which offer large potential revenue streams from the energy, capacity, and ancillary services markets as well as greater long-term revenue certainty given the durability of wholesale markets. In comparison, individual utility-level VPP programs tend to have short-term contracts (1-2 years), which creates greater revenue uncertainty for aggregators.⁵⁷

See [Appendix E.i.](#) for a detailed case study on how Leap aggregates demand response to participate in the CAISO market.

While total revenue potential across wholesale markets is large, each ISO/RTO has a unique set of rules and processes that require deep expertise to navigate, creating barriers for new entrants. As a result, participating in wholesale markets may provide lower levels of compensation than current utility-led VPPs receive today. Additionally, most ISO/RTOs only allow large-load demand response and do not yet allow DERs that store or generate energy (e.g., distributed storage and solar PV) to export power to the grid. This limits the value that DERs can bring to wholesale markets to a fraction of their technical functionality.

Streamlining wholesale market integration and allowing the full range of potential grid services from installed DERs could help address increasing capacity constraints that are causing price spikes and diminishing reserve margins across the U.S. For example, **PJM**, an RTO that coordinates wholesale electricity markets in all or parts of 13 states and the District of Columbia, held a capacity auction in summer of 2024 that resulted in final capacity prices nearly 10x higher than the previous year's auction.^{58, cviii, cix, cx, cxi}

57 Uncertainty around grid services revenue increases the cost of capital for industry actors investing in VPP participant recruitment and/or DER deployment. Longer-term contracts with greater revenue predictability can reduce the overall cost of VPP deployment, resulting in higher savings to pass on to customers.

58 In early 2024, PJM updated its capacity accreditation methodology to reflect the marginal contribution each resource can provide to system resource adequacy given the anticipated resource mix. As a result, many supply resources (including solar PV, gas, coal, hydropower, demand response) had lower capacity that could bid into the capacity market, resulting in lower capacity. Simultaneously, many existing power plants were forecasted to retire, further constraining supply and increasing PJM capacity prices.

As a result, PJM ratepayers will be responsible for \$14.7 billion in capacity costs for the 2025-2026 delivery year, as compared to \$2.2 billion for the 2024-2025 delivery year.^{cxii,cxiii} In early 2024, PJM updated its capacity accreditation methodology for all supply resources, including demand response. PJM's accreditation is based on PJM's existing requirement that DR resources be available for dispatch only between 10am-10pm during the summer and between 6am-9pm during the winter, even though DR resources could also perform outside these windows – effectively derating demand response because of PJM's rules rather than technological reality.

5.ii. Overview of FERC Order 2222

FERC Order 2222 has the potential to dramatically accelerate national action towards integrating VPPs into wholesale markets, which could maximize the value of DERs in restructured regions and help address rising affordability and reliability challenges to meet demand growth. Issued in September 2020, Order 2222 *requires* the six FERC-jurisdictional ISO/RTOs⁵⁹ to establish participation models that enable DER aggregations to participate in energy, capacity, and ancillary services wholesale markets.^{60,cxiv}

In issuing the Order, FERC recognized that a much wider range of DERs can provide wholesale market services, including those that export power and smaller individual assets.⁶¹ The Order is meant to offer a path to expand supply-side participation by DERs beyond demand response and place downward pressure on prices in markets with high demand and low supply. However, successful implementation of Order 2222 will require coordinated action from a broad range of stakeholders including ISO/RTOs, utilities, aggregators, regulators, and policymakers across the country.

5.iii. ISO/RTO Order 2222 compliance status

ISO/RTO compliance with FERC Order 2222 requirements has been varied: CAISO, NYISO, and ISO-NE are leading implementation while PJM, MISO, and SPP are seeking to implement much of their Order 2222 compliance proposals several years later. Although all six FERC-jurisdictional ISO/RTOs have filed compliance proposals with FERC, and FERC has issued orders on these filings, **CAISO** and **ISO-NE** are the only ISO/RTOs that have fully complied with the requirements of FERC Order 2222 as of December 2024.^{cxv,cxvi}

59 FERC does not have ratemaking jurisdiction with respect to ERCOT in Texas.

60 Specifically, Order 2222 requires each ISO/RTO to (a) develop tariff provisions that ensure that market rules facilitate the participation of DER aggregations, (b) allow DER aggregations to participate directly in ISO/RTO markets, and (c) establish DER aggregators as a type of market participant that can register DER aggregations.

61 FERC declined requiring ISO/RTOs to adopt minimum capacity requirements for individual distributed energy resources to participate in the markets, given those resources would only participate in the markets through a DER aggregation which would act as a single resource. However, some market operators have adopted minimums for individual DERs. For example, NYISO proposed a minimum capacity of 10 kW for each individual DER in any aggregation for a VPP to be eligible to participate, which would exclude many residential DER types.

Order 2222 compliance status

Issue Areas	CAISO	ISO-NE	NYISO	PJM	MISO	SPP
Metering and telemetry system requirements	In compliance	In compliance	Not yet in compliance	Not yet in compliance	Not yet in compliance	Not yet in compliance
Participation model	In compliance	In compliance	Not yet in compliance	In compliance	In compliance	Not yet in compliance
Double counting of services	In compliance	In compliance	In compliance	Not yet in compliance	In compliance	Not yet in compliance
Locational requirements	In compliance	In compliance	In compliance	In compliance	Not yet in compliance	Not yet in compliance
Role of distribution company	In compliance	In compliance	In compliance	Not yet in compliance	Not yet in compliance	Not yet in compliance
Ongoing operational coordination	In compliance	In compliance	In compliance	Not yet in compliance	Not yet in compliance	Not yet in compliance
Small utility opt-in	In compliance	In compliance	In compliance	In compliance	In compliance	In compliance
Interconnection	In compliance	In compliance	In compliance	In compliance	Not yet in compliance	Not yet in compliance
Definitions of DER and DER aggregator	In compliance	In compliance	In compliance	In compliance	In compliance	In compliance
Types of technologies	In compliance	In compliance	In compliance	In compliance	In compliance	Not yet in compliance
Allow a DER to serve as its own aggregator	In compliance	In compliance	In compliance	In compliance	In compliance	In compliance
Min and max size of aggregation	In compliance	In compliance	In compliance	In compliance	In compliance	In compliance
Min and Max size for DER in an aggregation	In compliance	In compliance	In compliance	In compliance	In compliance	Not yet in compliance
Distribution factors and bidding parameters	In compliance	In compliance	In compliance	In compliance	Not yet in compliance	Not yet in compliance
Information and data requirements	In compliance	In compliance	In compliance	Not yet in compliance	Not yet in compliance	In compliance
Role of RERRA	In compliance	In compliance	In compliance	In compliance	Not yet in compliance	Not yet in compliance
Modifications to list of resources in aggregation	In compliance	In compliance	In compliance	In compliance	In compliance	In compliance
Market participation agreements	In compliance	In compliance	In compliance	In compliance	Not yet in compliance	Not yet in compliance
Demand response opt-out	In compliance	In compliance	In compliance	In compliance	In compliance	Not yet in compliance

Legend

In compliance Not yet in compliance

Source: Lawrence Berkeley National Lab DER Participation in Wholesale Markets Report, FERC filings

ISO/RTO compliance plans exhibit individualized, disparate approaches to DER wholesale market participation, resulting in a patchwork of rules and requirements that make it difficult for aggregators to scale across jurisdictions. Order 2222 did not provide a technical implementation roadmap, leaving it up to the ISO/RTOs to make their own decisions on VPP integration standards and protocols. Market operators are taking different approaches to compliance with varying rules, baselining methodologies, grid services definitions, and operational protocols. For example, **ISO New England** requires telemetry readings to be actual data for all assets while **PJM** allows telemetry readings to be calculated based on a sample of DERs.^{cxvii}

While some variation is expected given varying market conditions and needs, the degree of variation across regions is introducing delays in scaling up proven VPP models nationally to address near-term grid needs. For example, one industry analysis estimated that standardizing ISO/RTO metering and settlement approaches alone could create \$75B in savings due to reductions in data computing, storage, and management costs.^{cxviii}

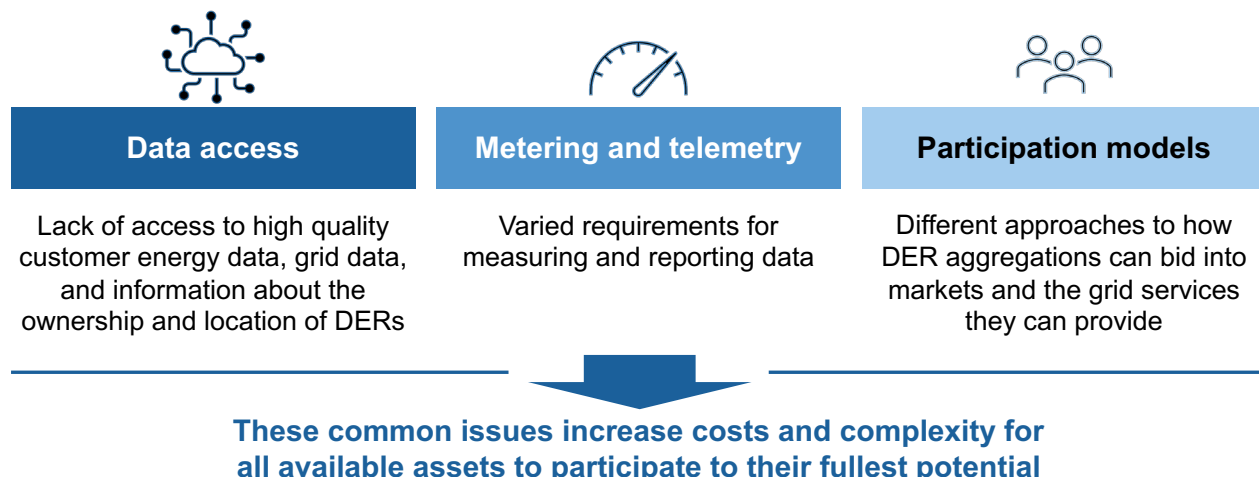
Even in achieving Order 2222 compliance, ISO/RTOs could still limit VPP participation. For example, separate from its compliance with Order 2222, **NYISO** proposed a minimum capacity of 10 kW for each individual DER in any aggregation for a VPP to be eligible to participate, which would exclude many residential DER types.^{62,cxix} In California, even though **CAISO** has reached full compliance with Order 2222, **California Public Utilities Commission (CPUC)** does not recognize aggregations of DERs as qualified to provide resource adequacy, which is one of the major barriers for aggregators that want to directly participate in CAISO's resource adequacy construct.^{cxx}

62 In a September 2022 presentation, NYISO wrote, 'Given the NYISO's current technical resources and capabilities for initial DER deployment, allowing small (<10 kW) DER will require a substantial amount of additional manual work to complete tasks that are core to the timely participation of DERs.'

5.iv. Common challenges for integration into wholesale markets

VPP aggregators, ISO/RTOs, and utilities have noted three key issue areas that impact VPP integration into wholesale markets.^{cxxi}

Common issue areas to VPP integration into wholesale markets



- 1. Data access: Lack of access to high quality customer energy data, grid data, and information about the ownership and location of DERs limits aggregators' ability to establish and operate reliable VPPs that can participate in wholesale markets.**

Hourly and daily customer energy data is required for aggregators to complete wholesale market processes such as receiving customer consent to access data, calculating baselines and performance, and implementing settlement procedures. Most ISO/RTOs require VPP aggregators to use customer energy data from utility-owned meters, yet most utilities limit third-party data access, citing cybersecurity and data privacy concerns. Aggregators must follow varied and often lengthy processes to access customer energy data from utilities, which makes it difficult to know if, when, and in what format data will be shared across utilities even within the same ISO/RTO region.⁶³ Aggregators have had limited success in improving data access rules despite filing complaints and requesting improvements.^{64,cxxii,cxxiii} FERC has stated that customer usage data access is not within FERC's jurisdiction, leaving it to individual ISO/RTOs, states, tribes, or utilities to determine data access policies.^{cxxiv}

- 2. Metering and telemetry: Each ISO/RTO is developing individual frameworks to determine how to measure and report DER energy data, which increases the complexity and cost for aggregators to deploy and scale VPPs.**

As relates to Order 2222, "metering" refers to the rules that determine how DER aggregations measure energy injection and withdrawal; "telemetry" refers to how aggregations report the real-time data needed to provide grid services. ISO/RTOs require VPPs to use meter data for planning, operations, and settlement purposes. While the technological capabilities exist to capture this data, such as with

⁶³ For example, sometimes aggregators must email utilities with a form requesting a customer's data. In other cases, utilities have portals where aggregators can access the data directly.

⁶⁴ For example, CPower, an aggregator who works closely with utilities and supports utility programs, brought a complaint before FERC (EL23-104) showing that lack of data access limits demand response participation in wholesale markets. CPower argued that PJM rules limit the participation of aggregators by refusing to allow statistical sampling for measurement purposes. If statistical sampling were allowed, CPower could measure behavior of a representative sample of customers in their aggregation who have advanced metering infrastructure (AMI) and extrapolate the performance to the full aggregation. PJM, however, requires measurement from every individual meter for interval metered customers, which dramatically reduces CPower's capacity contribution because utilities in the region block access to data from many meters where customers could otherwise participate. FERC denied the complaint and noted that customer usage data access is not within their jurisdiction. Additional detail provided here: [CPower Statement Regarding September 19 FERC Decision - CPower Energy; Order Denying Complaint re Enerwise Global Technologies, LLC v. PJM Interconnection, L.L.C. under EL23-104](#)

advanced metering infrastructure (AMI)⁶⁵ or metering manufactured into devices, most utilities do not measure demand at this granularity and ISO/RTO telemetry requirements can be too costly or complex for certain asset types.

For Order 2222 compliance, FERC did not provide strict guardrails for either metering or telemetry, approving each ISO/RTO's framework so long as they justify how their requirements are just and reasonable and do not pose an unnecessary and undue barrier to individual DERs joining an aggregation. This has led to a wide variety of approaches that are seeking to balance data granularity with the costs of reporting high frequency data. For example, **NYISO** requires six-second telemetry for every DER asset that is at least 100 kW, regardless of the service provided. **PJM** allows one-minute scans for resources that do not provide regulation services and entirely exempts DERs under 10 MW from telemetry reporting.^{CXXV}

3. Participation models: ISO/RTOs are taking different approaches to develop market participation rules that define how DER aggregations are allowed to bid into wholesale markets and the grid services they can provide, making it difficult for aggregators to replicate similar models across markets.

ISO/RTOs have multiple choices in determining market participation rules that define how aggregations bid into the market. An ISO/RTO can choose between requiring aggregators to comply with rules already established for existing supply resources (e.g., applying rules for utility-scale batteries to aggregations of residential behind-the-meter batteries), creating new participation models specifically for DER aggregations, or using a hybrid approach.

There are trade-offs between these approaches. For example, leveraging models for existing supply resources could avoid slow, expensive processes to create new participation rules, but may restrict participation from aggregations that have multiple types of DERs. Creating a single, new participation model for DER aggregations could simplify aggregator choice on how to participate in a wholesale market but may require all types of DER aggregations to comply with the same rules (e.g., battery-only aggregation may have the same rules as an aggregation with batteries, thermostats, and commercial & industrial loads).⁶⁶ Offering a hybrid approach, as **NYISO** and **ISO-NE** are currently suggesting, allows aggregators to choose the option that is highest value to their business model.^{CXXVI}

Outside these three challenges, additional issue areas include how to coordinate and compensate dual participation of DERs across wholesale and retail markets⁶⁷ (i.e., avoiding 'double counting' for the same service), locational requirements on aggregating DERs across eligible pricing nodes, and ongoing coordination between distribution utilities, market operators, and aggregators.

⁶⁵ AMI or 'smart meters' are used to measure a customer's energy consumption during set time intervals. AMI includes technologies to measure and communicate energy use and other data and notifications at intervals that are granular enough to support grid and market operations.

⁶⁶ ISO-NE created multiple DER aggregation participation models to address the drawbacks of creating only one new participation model that would apply to every type of DER aggregation.

⁶⁷ There is still a role for utilities to play to compensate VPPs for distribution benefits separately and in addition to wholesale market compensation to cover the full value stack of potential services. A VPP that delivers benefits to the transmission system and to the distribution system (even if during the same event) can be fairly compensated for both.

5.v. Supportive actions for VPP integration into wholesale markets

ISO/RTOs, state regulators, utilities, and aggregators can collaborate to streamline learnings and converge on comparable approaches that address common issues, enabling VPPs to better meet near-term grid capacity needs at lower costs for ratepayers.

There are multiple solutions available globally that could be adopted to support VPP integration into wholesale markets.

Case Study: Australian Energy Market Operator (AEMO)

Australian Energy Market Operator established a centralized, standardized DER registry to provide visibility to DER specifications and location to eligible entities.



- ▶ In 2020, the **Australian Energy Market Operator (AEMO)** established a [centralized DER registry](#) to better manage the grid, improve system reliability as the grid becomes more decentralized, and deliver energy at a more affordable price.
- ▶ The register provides a common, standardized information fact base with visibility to DER specifications (e.g., type, capabilities, resource ownership) and location.
- ▶ Customers, AEMO, distribution utilities, DER industry, and other third parties (such as emergency services) can access the register.
- ▶ Entities are required to provide data in certain formats and timelines; for example, utilities are required to provide DER information in accordance with the DER Register Information Guidelines under the National Electricity Rules to ensure standardization, and DER installers are required to submit data within 20 days of installation.^{68,cxxvii,cxxviii,cxxix}

Case Study: Ontario Independent Electricity System Operator (IESO)

Ontario Independent Electricity System Operator (IESO) created market-wide standards for meter registration to standardize data collection and reduce IT costs



- ▶ **Ontario IESO** has established [market-wide standards for meter registration](#) across numerous distribution utilities and 5 million smart meters.^{cxxx}
- ▶ Market rules require that each metering installation used for settlement purposes is on a list of pre-approved meters established by IESO that meet specific performance standards (e.g., accuracy, security).
- ▶ Establishing a market-wide approach to metering simplifies and standardizes data collection while reducing IT costs to develop, manage, and protect the database.
- ▶ This spurred additional engagement with various grid stakeholders to expand third-party access to this database, including for demand response aggregators.^{cxxxi,69}

See [Appendix E](#) for 6 case studies on actions ISO/RTOs have been taking domestically and internationally to integrate VPPs into wholesale markets ([Appendix E.iii.](#)), 6 additional resources ([Appendix E.iv.](#)) and 3 supportive DOE programs ([Appendix E.v.](#)).

⁶⁸ The Australian Energy Market Commission made a rule obligating AEMO to establish this register in the National Electricity Market in September 2018. AEMO engaged with a wide range of partners, including utilities and industry groups, to design the register and align on the corresponding data sets and data collection processes.

⁶⁹ Another example is ConnectedSolutions, which has metering authority across multiple utilities in Massachusetts. Common program design across utilities enables standardization of data access, dispatch, monitoring and verification, and DERMS while providing economies of scale for enrollment.

In parallel with ISO/RTO implementation of Order 2222, state policymakers and regulators can act to build enabling VPP regulations and policies that further integrate VPPs into wholesale markets.

Example actions include:

- **Lifting state-level ‘opt outs’ on Order 719:** FERC Order 719 was introduced in 2008 to allow demand response to participate in wholesale markets alongside traditional supply-side resources.^{xxxxii} However, states were allowed to ‘opt out,’ by prohibiting third-party aggregators from directly contracting with customers. These ‘opt outs’ have greatly limited DER market participation in these states. **Missouri PSC** ruled to partially lift its FERC Order 719 opt out in October 2023 by allowing energy customers above 100 kW (commercial & industrial loads) to enter **MISO’s** demand response market.^{xxxxiii} By starting with commercial and industrial loads, Missouri state regulators and utilities could test and learn to inform more complex future policies and VPP integration approaches. **Michigan** and **Wisconsin** have also partially lifted their initial ‘opt out’ of Order 719. Ten states still have ‘opt outs’ in place for Order 719.⁷⁰
- **Determining the state regulator’s role in Order 2222 implementation:** **Pennsylvania PUC** issued an [Advanced Notice of Proposed Rulemaking](#) in February 2024 to investigate the PUC’s role in Order 2222 implementation. Topics identified for stakeholder input included DER interconnection rules, metering requirements, data sharing protocols, and cost allocation processes.^{xxxxiv}
- **Requiring utilities to meet data sharing standards:** **Connecticut PUC** created a [Data Access and Privacy Framework](#) to clarify data requirements for IOUs deploying AMI, including data sharing expectations with third-party aggregators.^{xxxxv} In response, **Eversource** agreed to adopt Green Button Connect to enable third-party data access.^{xxxxvi} Similarly, **Rhode Island PUC** is requiring **Narragansett Electric Company** to [submit a plan](#) about data access (including for VPPs) as part of the utility’s planned investment into AMI.^{xxxxvii}

70 The ten states that continue to fully opt out of Order 719 are Arkansas, Indiana, Iowa, Kentucky, Louisiana, Minnesota, Mississippi, Montana, North Dakota, and South Dakota.